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COURSE NAME – Operation System
COURSE CODE – CSA0408

Lab Observations

EXPERIMENT - 16

Develop a C program for implementing random access file for processing the employee details

PROGRAM:

```
#include <stdio.h>
#include <stdlib.h>

struct Employee
{
    int empId;
    char empName[50];
    float empSalary;
};

int main()
{
    FILE *filePtr;
    struct Employee emp;
    int choice;

    filePtr = fopen("employee.dat", "rb+");

    if (filePtr == NULL)
        filePtr = fopen("employee.dat", "wb+");

    if (filePtr == NULL)
    {
        printf("Error opening file.\n");
        return 1;
    }
```

```
do
{
    printf("\n1. Add Employee");
    printf("\n2. Display Employee Details");
    printf("\n3. Update Employee Details");
    printf("\n4. Exit");
    printf("\nEnter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            printf("Enter Employee ID: ");
            scanf("%d", &emp.empId);

            printf("Enter Employee Name: ");
            scanf("%49s", emp.empName);

            printf("Enter Employee Salary: ");
            scanf("%f", &emp.empSalary);

            fseek(filePtr,
                  (emp.empId - 1) * sizeof(struct Employee),
                  SEEK_SET);

            fwrite(&emp, sizeof(struct Employee), 1, filePtr);
            printf("Employee added successfully.\n");
            break;

        case 2:
            printf("Enter Employee ID to display: ");
            scanf("%d", &emp.empId);

            fseek(filePtr,
                  (emp.empId - 1) * sizeof(struct Employee),
                  SEEK_SET);

            if (fread(&emp, sizeof(struct Employee), 1, filePtr) == 1
                && emp.empId > 0)
            {
                printf("Employee ID: %d\n", emp.empId);
                printf("Employee Name: %s\n", emp.empName);
                printf("Employee Salary: %.2f\n", emp.empSalary);
```

```
        }
        else
        {
            printf("Employee not found.\n");
        }
        break;

    case 3:
        printf("Enter Employee ID to update: ");
        scanf("%d", &emp.empId);

        fseek(filePtr,
              (emp.empId - 1) * sizeof(struct Employee),
              SEEK_SET);

        if (fread(&emp, sizeof(struct Employee), 1, filePtr) == 1
            && emp.empId > 0)
        {
            printf("Enter new Employee Name: ");
            scanf("%49s", emp.empName);

            printf("Enter new Employee Salary: ");
            scanf("%f", &emp.empSalary);

            fseek(filePtr,
                  (emp.empId - 1) * sizeof(struct Employee),
                  SEEK_SET);

            fwrite(&emp, sizeof(struct Employee), 1, filePtr);
            printf("Employee updated successfully.\n");
        }
        else
        {
            printf("Employee not found.\n");
        }
        break;

    case 4:
        printf("Exiting program.\n");
        break;

    default:
        printf("Invalid choice.\n");
```

```
        }

    } while (choice != 4);

    fclose(filePtr);
    return 0;
}
```

OUTPUT:

